

WineQT.csv – Classification Data Quality Report

Date: February 22, 2026

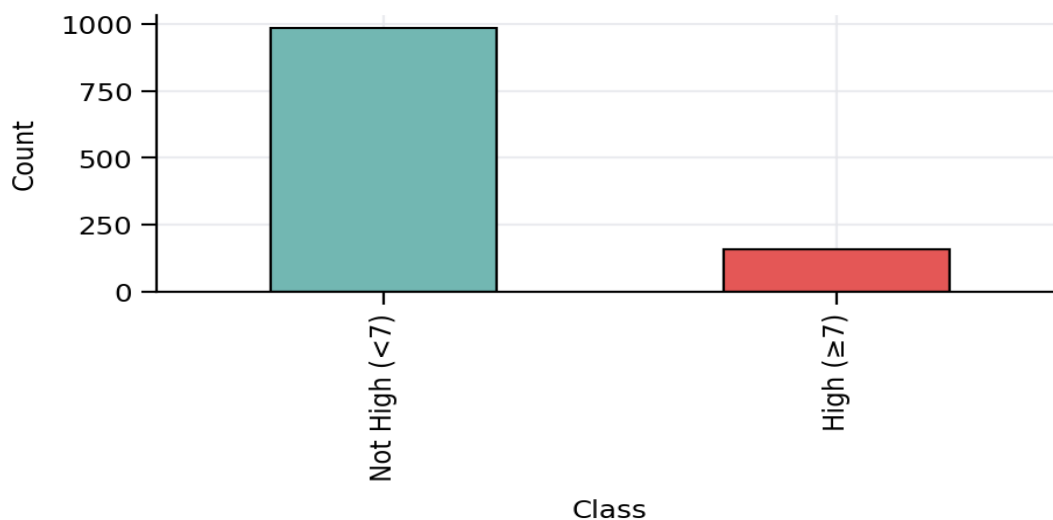
Observations: 1,143 | Variables: 13

Classification Target: High quality (quality ≥ 7). Class counts → Not High: 984 | High: 159

Executive Summary (one page)

- No missing cells; 0 duplicate rows.
- Quality is imbalanced (mid-range dominates); use stratified splits and class weighting when modeling.
- Multiple variables show right-tailed outliers; use robust scaling or IQR-winsorization (see table).
- Rule checks found potential plausibility issues (counts shown below); investigate flagged rows (CSV provided).
- Top correlates with quality (Spearman) are highlighted; monitor multicollinearity among acidity/SO₂ blocks.

Class Distribution (High Quality = quality ≥ 7)



Recommended next steps: (1) Robust scaling + optional winsorization; (2) Stratified train/validation split; (3) Tree/boosting models or regularized linear models with collinearity control; (4) Monitor feature importance and SHAP for directionality.

Variable Summary & IQR/Winsorization Recommendations

Variable	Q1	Q3	IQR	Lower Outliers	Upper Outliers	Outlier %	Skewness	Recommendation
pH	7.1	9.1	2	0	44	3.85%	1.04	Winsorize to [4.1, 12.1] (IQR capping)
Volatile acidity	0.393	0.64	0.247	0	14	1.22%	0.68	Winsorize to [0.0213, 1.01] (IQR capping)
Citric acid	0.09	0.42	0.33	0	1	0.09%	0.37	No capping; prefer robust scaling
Residual sugar	1.9	2.6	0.7	0	110	9.62%	4.36	Winsorize to [0.85, 3.65] (IQR capping)
Chlorides	0.07	0.09	0.02	8	69	6.74%	6.02	Winsorize to [0.04, 0.12] (IQR capping)
Free sulfur dioxide	7	21	14	0	18	1.57%	1.23	Winsorize to [-14, 42] (IQR capping)
Total sulfur dioxide	21	61	40	0	40	3.50%	1.66	Winsorize to [-39, 121] (IQR capping)
Density	0.996	0.998	0.00228	16	20	3.15%	0.10	Winsorize to [0.992, 1] (IQR capping)
Alcohol	3.21	3.4	0.195	8	12	1.75%	0.22	Winsorize to [2.91, 3.69] (IQR capping)
Phosphates	0.55	0.73	0.18	0	43	3.76%	2.49	Winsorize to [0.28, 1] (IQR capping)
Alcohol	9.5	11.1	1.6	0	12	1.05%	0.86	Winsorize to [7.1, 13.5] (IQR capping)
Acidity	5	6	1	6	16	1.92%	0.29	Winsorize to [3.5, 7.5] (IQR capping)

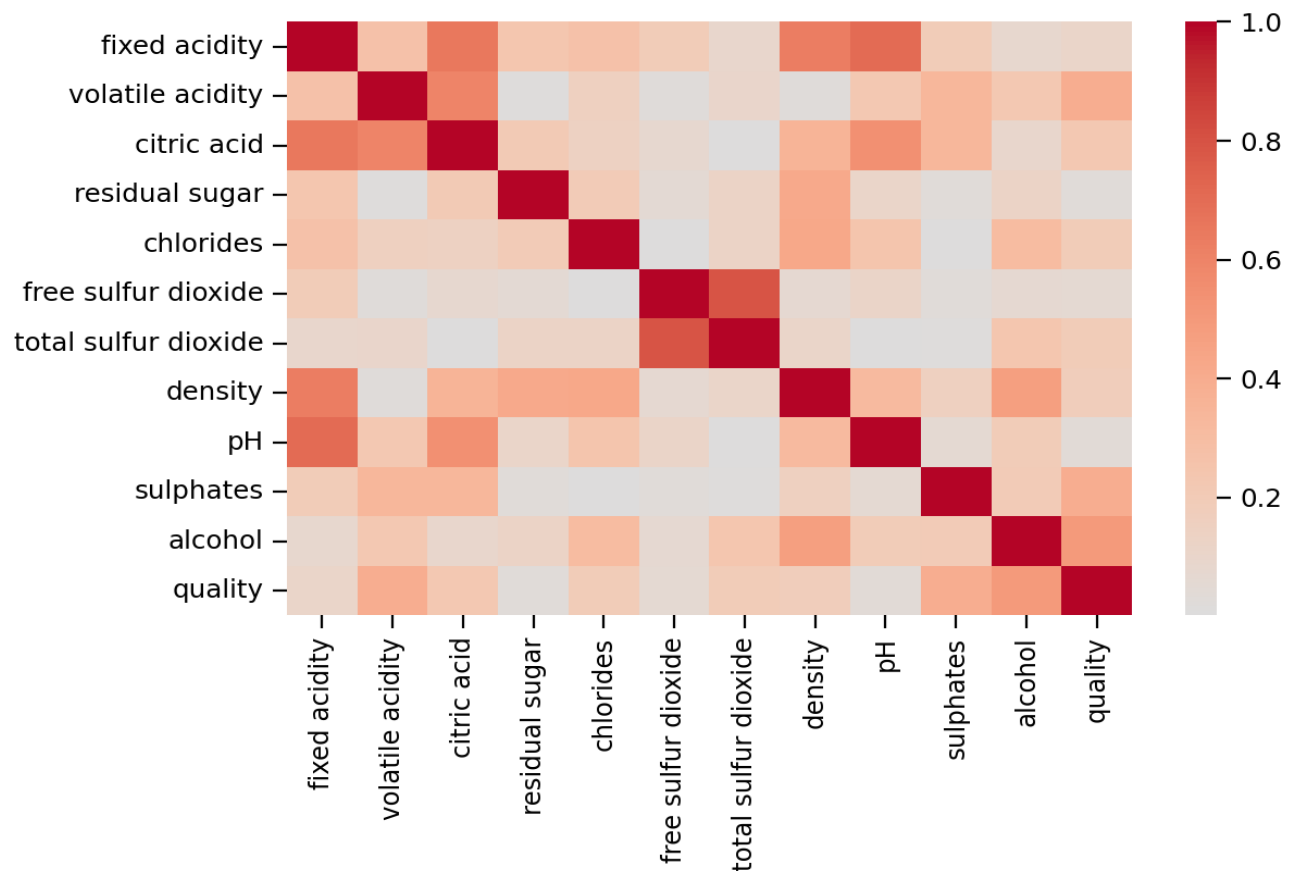
Plausibility & Business-Rule Checks

Rule	Violations
total SO2 < free SO2	0
pH not in [2, 4]	2
density not in [0.98, 1.04]	0
alcohol not in [5, 23] (%)	0
residual sugar < 0	0
chlorides < 0	0
sulphates < 0	0
citric acid < 0	0
volatile acidity < 0	0
fixed acidity < 0	0
free sulfur dioxide < 0	0
total sulfur dioxide < 0	0

Feature-Selection Diagnostics

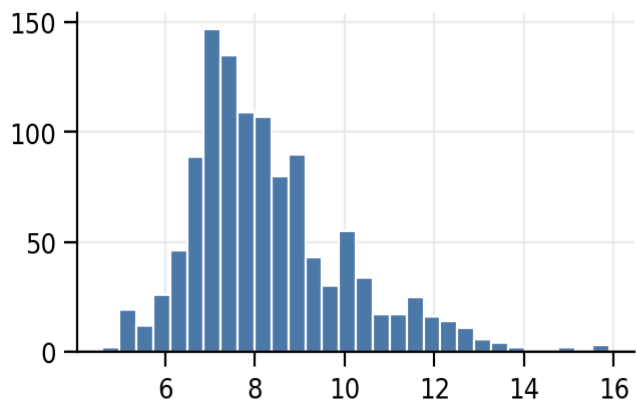
Check	Details
Constant columns	None
Highly correlated feature pairs ($ r \geq 0.95$, Spearman)	None
Top 5 correlations vs quality (Spearman)	quality: 1.00 alcohol: 0.50 volatile acidity: -0.40 sulphates: 0.39 citric acid: 0.22
Leakage flags	No obvious leakage detected

Correlation Heatmap (Numeric Variables)

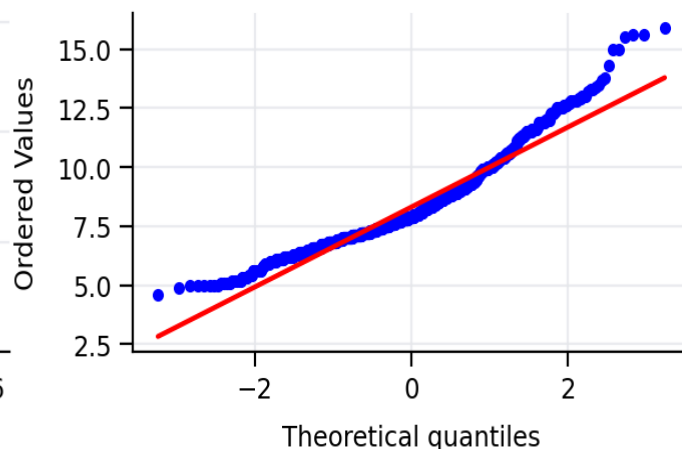


Per-Variable Histograms & Q-Q Plots

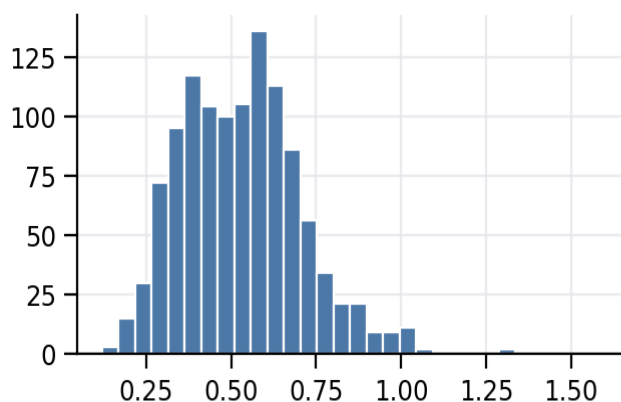
fixed acidity - Histogram



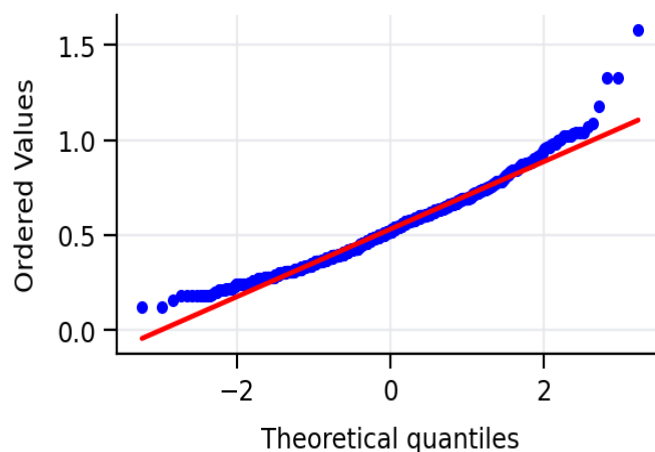
fixed acidity - Q-Q Plot



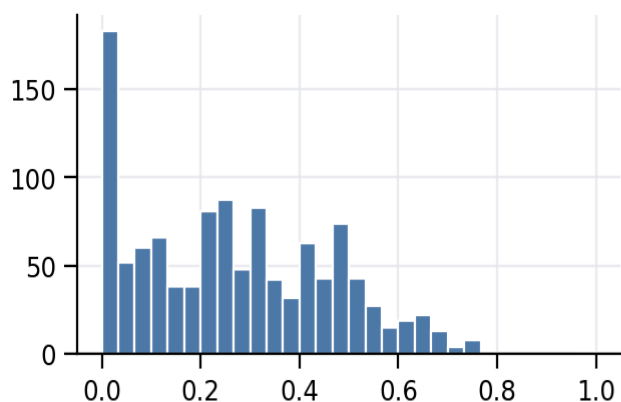
volatile acidity - Histogram



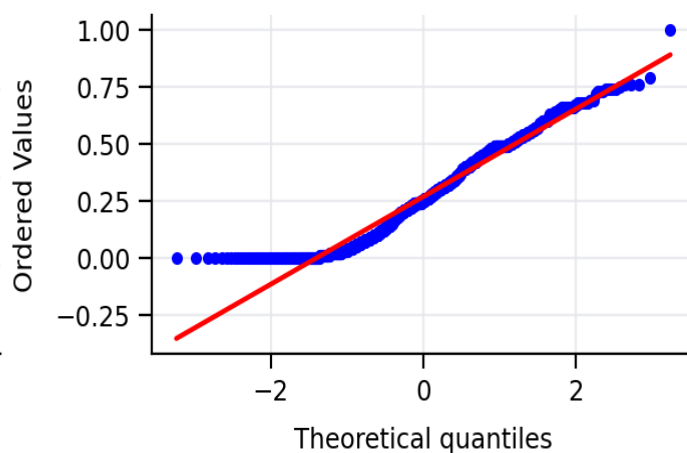
volatile acidity - Q-Q Plot



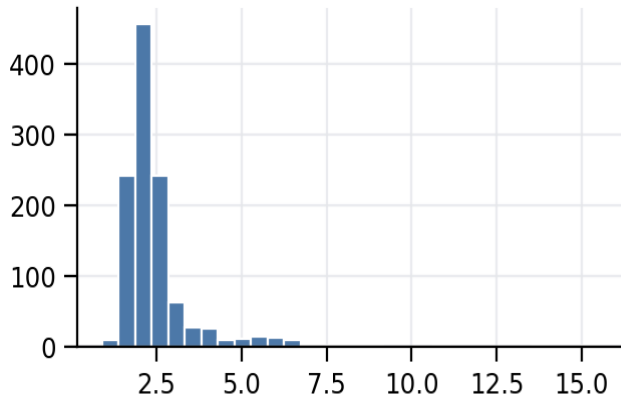
citric acid - Histogram



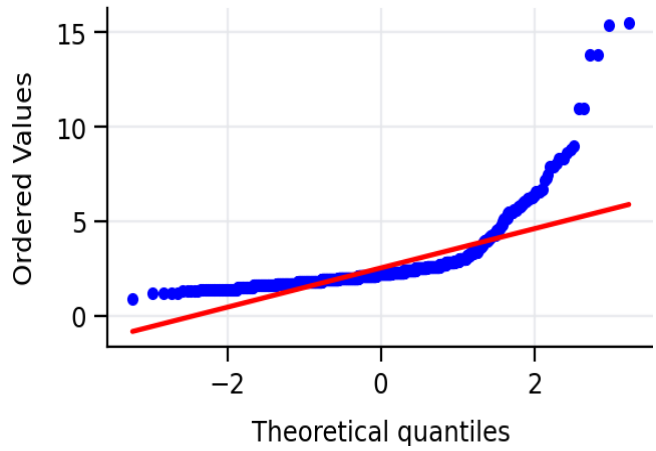
citric acid - Q-Q Plot



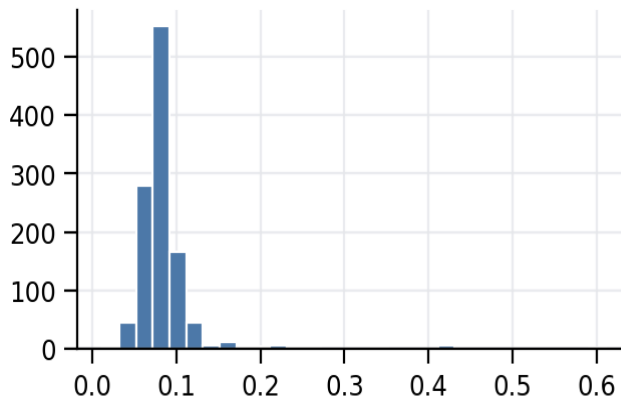
residual sugar - Histogram



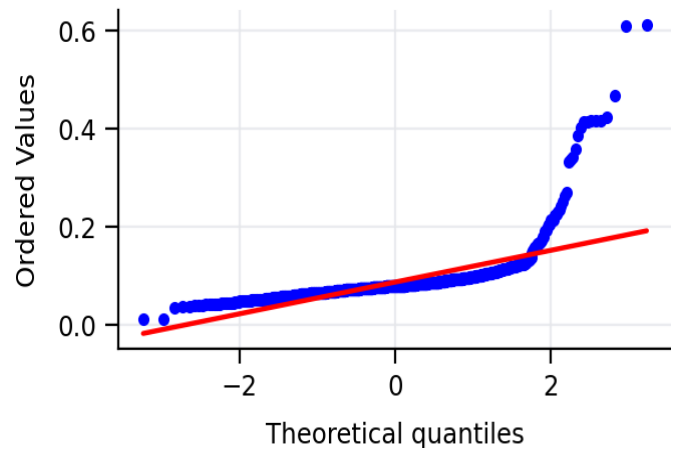
residual sugar - Q-Q Plot



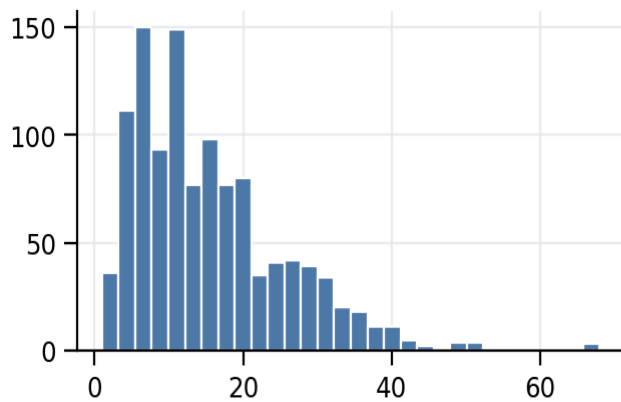
chlorides - Histogram



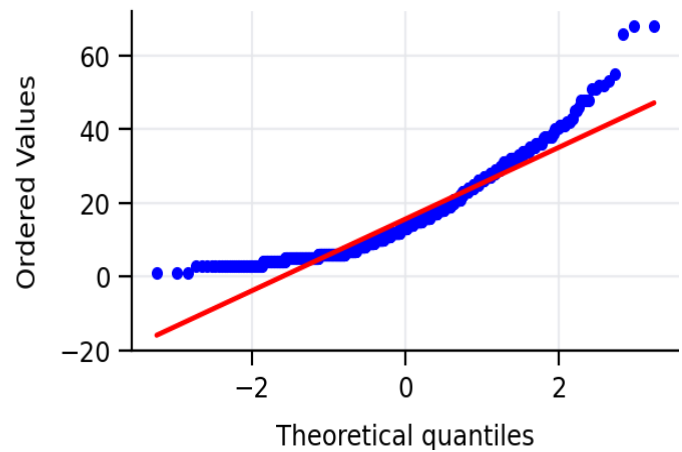
chlorides - Q-Q Plot



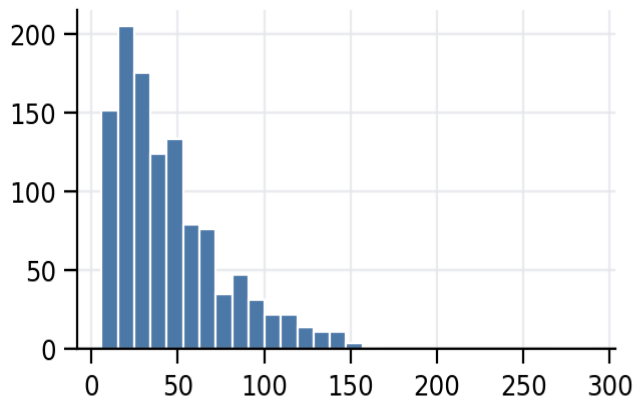
free sulfur dioxide - Histogram



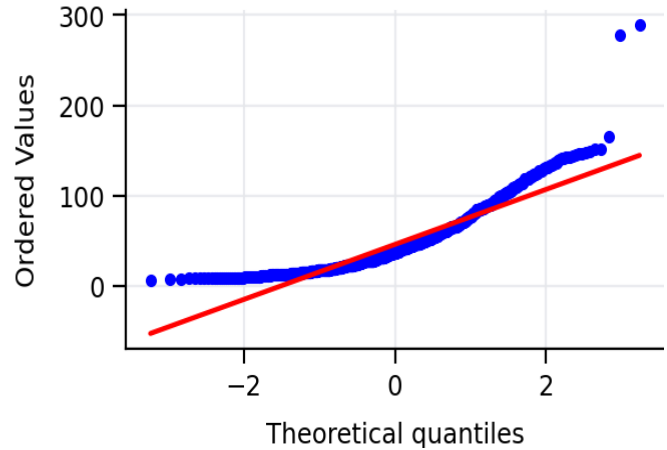
free sulfur dioxide - Q-Q Plot



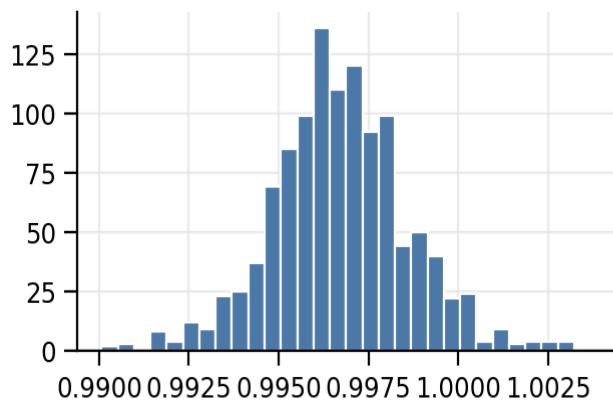
total sulfur dioxide - Histogram



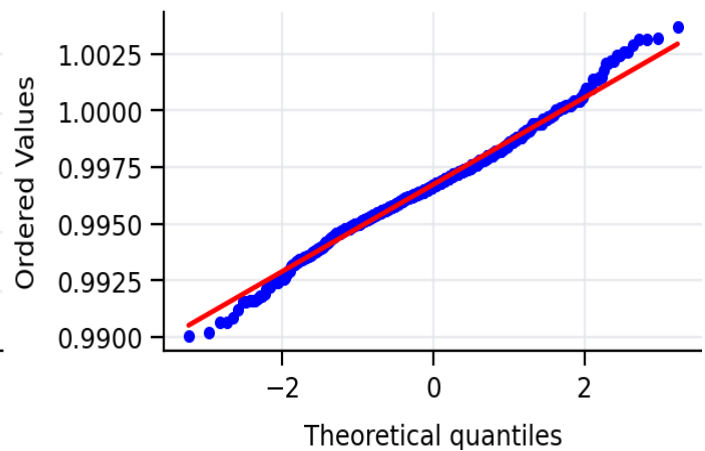
total sulfur dioxide - Q-Q Plot



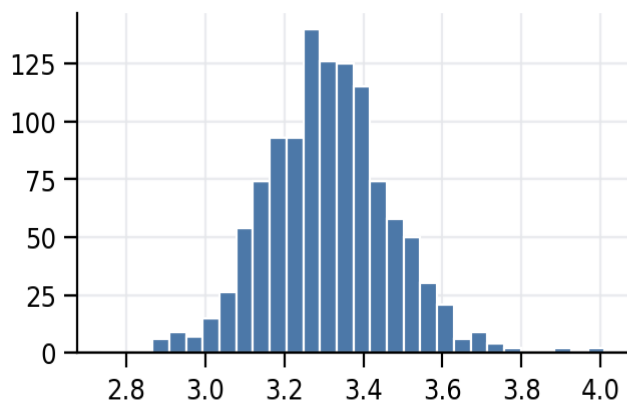
density - Histogram



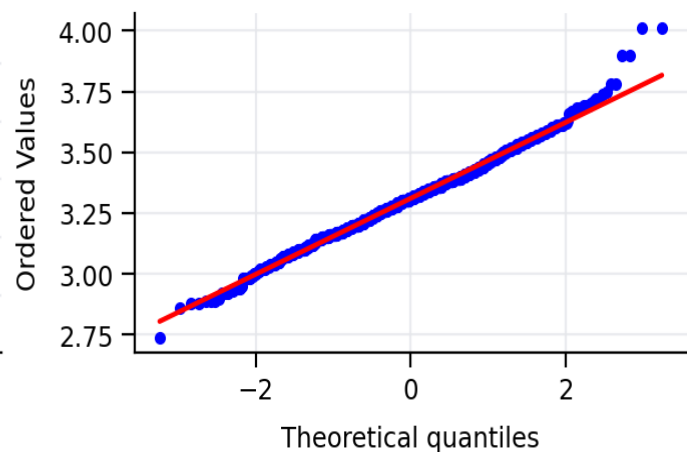
density - Q-Q Plot



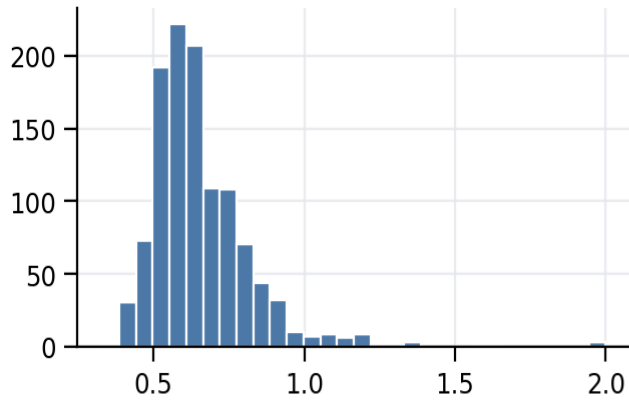
pH - Histogram



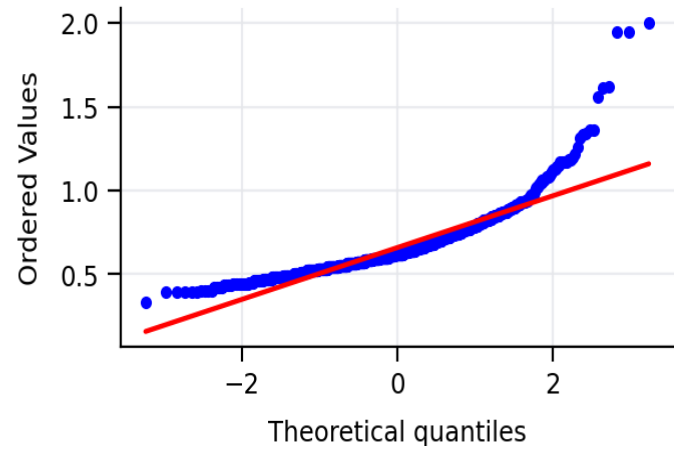
pH - Q-Q Plot



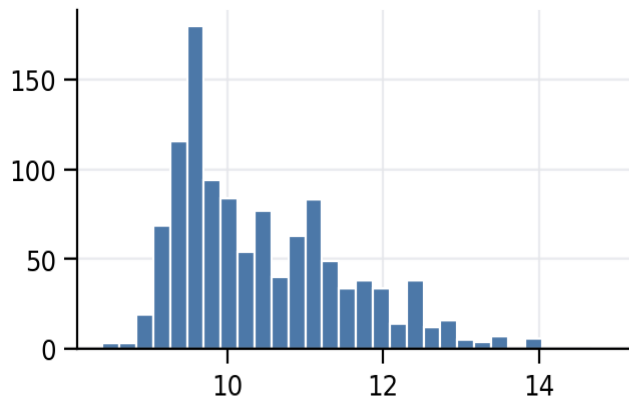
sulphates - Histogram



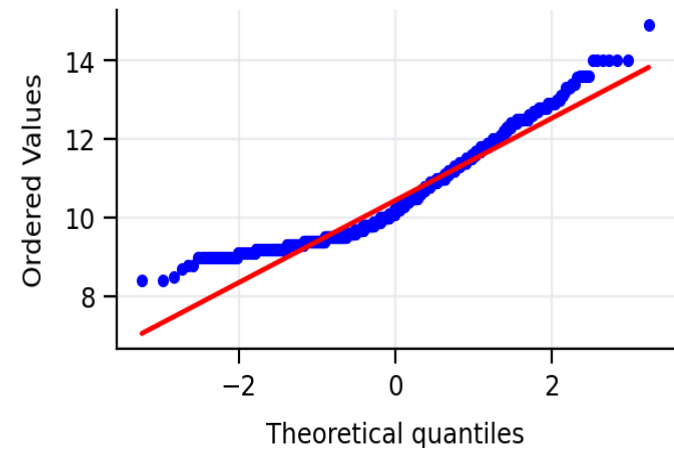
sulphates - Q-Q Plot



alcohol - Histogram



alcohol - Q-Q Plot



Appendix – Rows Violating Plausibility Rules

Total violating rows: 2. Showing up to first 50 rows.

	Id	quality	pH	density	alcohol	free sulfur dioxide	total sulfur dioxide	residual s
	1316	1316	6	4.01	0.99258	12.5	16.0	46.0
	1321	1321	6	4.01	0.99258	12.5	16.0	46.0

Full CSV of violating rows saved as: WineQT_rule_violations.csv